Compressed Sensing for Efficient Fidelity Estimation of GHZ States

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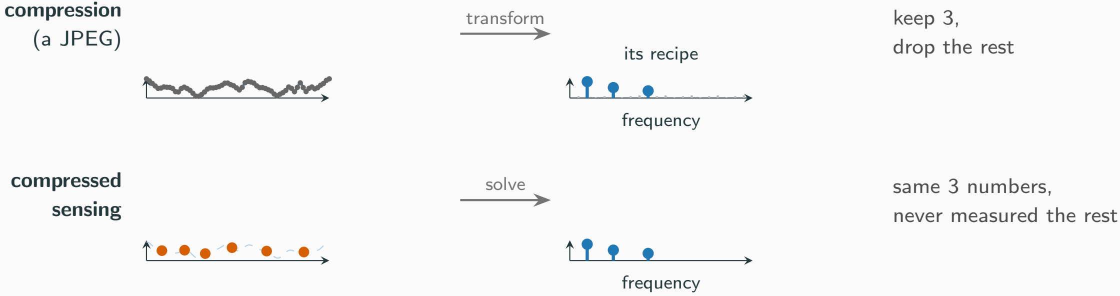
Farrokh Labib[1] David Nicholaeff[2] **Vincent Russo**[1] William J. Zeng[1,3]

arXiv:2604.27824

vprusso.github.io

[1]Unitary Foundation [2]New Mexico Consortium [3]Quantonation

What compressed sensing is



- The right-hand chart is the signal's recipe: the signal rewritten as a sum of simple waves, one entry per wave. Natural signals have nearly empty recipes: a JPEG keeps the few big entries and looks identical, but it measures everything first.
- Compressed sensing pays for the information, not the signal length: few random samples, solve for the big entries. Ours is the expensive case: each measurement is a quantum circuit.

GHZ states as a hardware benchmark

$$|\mathrm{GHZ}\rangle = \frac{1}{\sqrt{2}}\left(|0\rangle^{\otimes N} + |1\rangle^{\otimes N}\right)$$

secret sharing | metrology | error correction

120 qubits

largest GHZ state to date
(IBM, 2025; $F > 0.5$)[a]

1 bit flip

is enough to destroy
the global coherence

[a] Javadi-Abhari, Ali, et al. "Big cats: entanglement in 120 qubits and beyond." arXiv:2510.09520 (2025).

This talk: verify GHZ states with $\log N$ measurement circuits instead of $2N$, and detect errors during preparation on trapped-ion hardware.

Fidelity is four matrix elements

$$F(\rho) = \langle \mathrm{GHZ}|\rho|\mathrm{GHZ}\rangle = \frac{1}{2}\Big(\underbrace{\rho_{\mathbf{0},\mathbf{0}} + \rho_{\mathbf{1},\mathbf{1}}}_{\text{population } P} + \underbrace{\rho_{\mathbf{0},\mathbf{1}} + \rho_{\mathbf{1},\mathbf{0}}}_{\text{coherence term}}\Big)$$

$$\begin{array}{c} \langle 0\cdots 0| \\ \\ \\ \langle 1\cdots 1| \end{array} \begin{bmatrix} 0.5 & 0 & 0 & 0.5 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0.5 & 0 & 0 & 0.5 \end{bmatrix}$$

GHZ state: $F = 1$

$$\begin{bmatrix} 0.5 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.5 \end{bmatrix}$$

classical mixture: $F = \frac{1}{2}$

$$\begin{bmatrix} 0.5 & 0 & 0 & 0.3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0.3 & 0 & 0 & 0.5 \end{bmatrix}$$

noisy state: $F = 0.8$

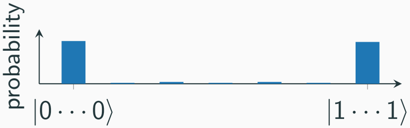
Diagonal entries are outcome probabilities; the corner off-diagonals measure the superposition itself.

A coin flip scores exactly $F = 1/2$. Everything above $1/2$ lives in the orange corners: C is the number to measure.

Fidelity from parity oscillations

Population P

chance of reading all-0s or all-1s

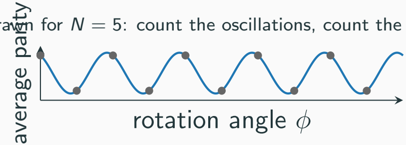


Z-basis counts: 1 circuit

Coherence C

rotate every qubit by ϕ, measure all; a shot's parity is $+1$ iff an even number read 1; average over shots

drawn for $N = 5$: count the oscillations, count the qubits

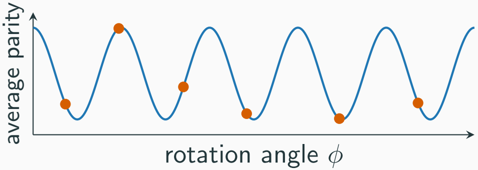


average parity $\mathcal{P}(\phi) = C\cos(N\phi + \theta)$. Resolving any frequency up to N: two samples per oscillation (Nyquist), the $2N = 10$ dots, each one circuit.

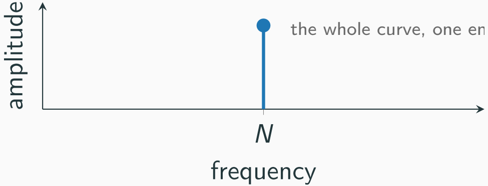
The fit returns amplitude $C \geq 0$ and phase θ; the fidelity is $F = (P + C\cos\theta)/2$.

At $N = 100$: 200 measurement circuits, just to verify the state.

The observation: the signal is sparse



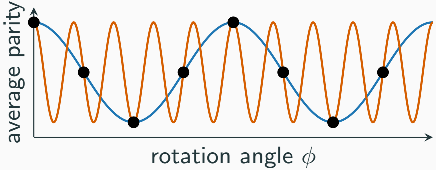
the curve's recipe: how much of each frequency



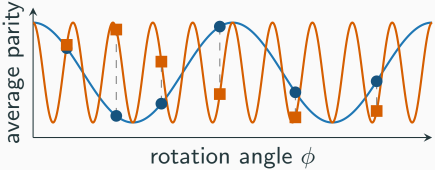
- For an ideal GHZ state, the average-parity curve is a single cosine at frequency N: one spike below.
- The bottom panel is the curve's recipe: how much of each candidate frequency it contains. Ours reads 0.9 at frequency N, zero elsewhere: 2 of ∼200 numbers (a cos and a sin amount). Sparse means the recipe is nearly empty.
- Recovering a sparse signal from few random samples is textbook compressed sensing: $M \propto \log N$ random angles (orange dots) replace the $2N$ equally spaced ones.

Why random beats regular

8 equally spaced angles: frequencies 2 and 10 give identical data



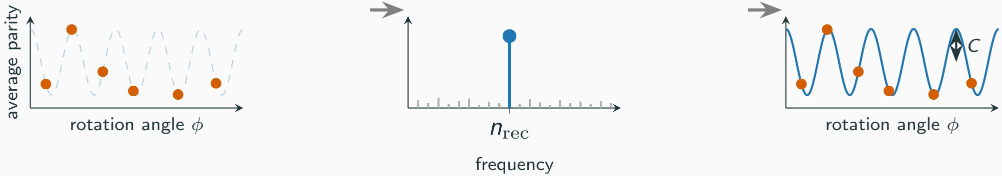
6 random angles: the same two frequencies disagree everywhere



- A regular grid has a period for impostors to hide behind: on $\phi_k = 2\pi k/8$, the collision is exact, $\cos(10\phi_k) = \cos(2\phi_k)$. Ruling this out for every frequency up to N is what costs $2N$.
- A random angle has no period: two distinct frequencies agree there with probability zero, so each further sample rules out a constant fraction of the wrong candidates: $\sim$ log(candidates) samples leave one survivor.

Recovery separates detection from estimation

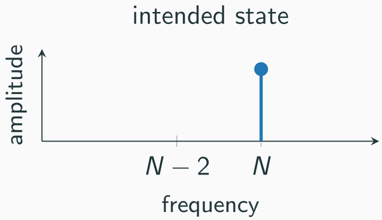
1. sample at M random angles 2. joint fit: penalty leaves one frequency 3. refit the survivor: C and θ

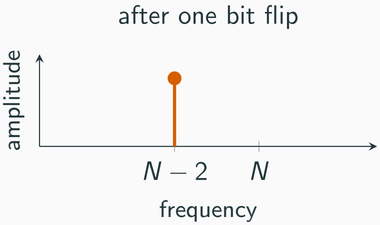


- Step 2 is the Lasso[1]: Fit all candidate frequencies together, with an L_1 penalty on the cosine and sine coefficients; weak candidates are driven exactly to zero.
- Why so few: each sample rules out a constant fraction of the wrong frequencies, so $M \propto \log N$ suffices; the refit then leaves only shot noise.

[1] Tibshirani, Robert. "Regression shrinkage and selection via the lasso." Journal of the Royal Statistical Society B 58.1 (1996).

The frequency doubles as a certificate





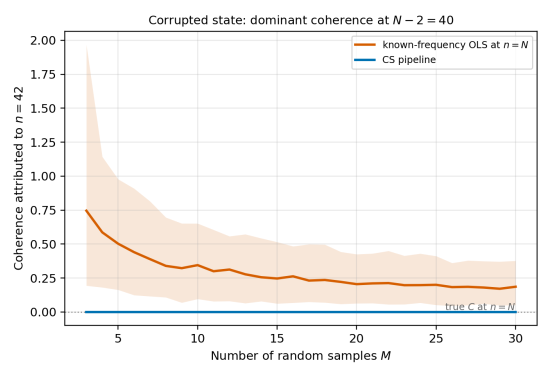
- Fixed-frequency fit at $n = N$: assigns amplitude to a frequency that is not there.
- Blind recovery: returns $n_{\mathrm{rec}} = N - 2$ and catches the failure.

$n_{\mathrm{rec}} = N$: the dominant coherence is the one an N-qubit GHZ state would have.

$n_{\mathrm{rec}} \neq N$: caught, not mis-certified.

With $F > 1/2$ this certifies genuine N-partite entanglement against the phase-optimized target. It does not rule out other errors or smaller competing coherences.

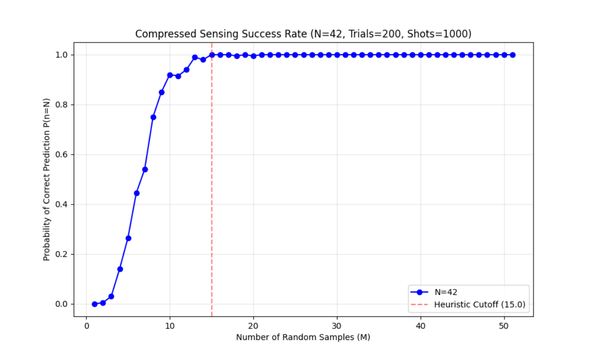
Fixed-frequency fitting produces false certificates



Synthetic corrupted state, emulator sampling: dominant coherence ($C = 0.8$) at $N-2$, zero coherence at N.

- Known-frequency OLS at $n = N$ reports median amplitude 0.75 at $M = 3$, still ≈ 0.19 at $M = 30$: a false-positive entanglement certificate.
- The CS pipeline recovers $n_{\mathrm{rec}} = N - 2$ and assigns zero coherence to N: the failure is caught.

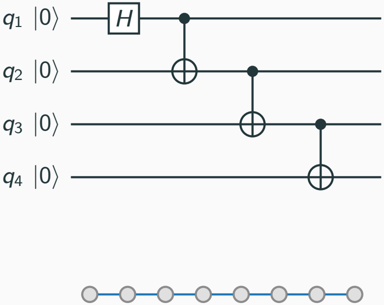
Recovery saturates below the working budget



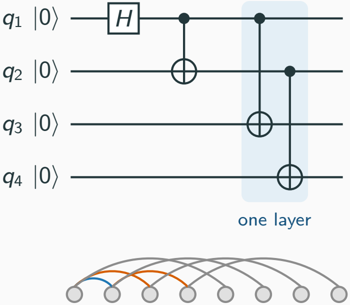
- How often does blind recovery return the right frequency? $P(n_{\mathrm{rec}} = N)$ at $N = 42$, 200 trials, 1000 shots per angle.
- Two free passes, then a sharp climb: success saturates by $M \approx 15$, below the $5 \ln N \approx 19$ we actually budget.
- That margin is why the hardware run used 19 angles at $N = 50$ instead of 100.

Preparing states worth verifying

Preparing a GHZ state: chain versus tree



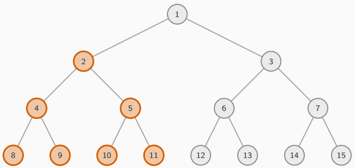
chain: depth $N-1$ (7 layers at $N=8$)



tree: depth $\log_2 N$ (3 layers at $N=8$)

- Same state, same gate count: only the CNOT schedule differs; every layer is an opportunity for noise, so shallower wins.
- A CNOT copies bit flips, so a fault near the root fans out to its whole subtree.
- Shallow preparation trades depth for high-weight errors.

How one bit flip spreads



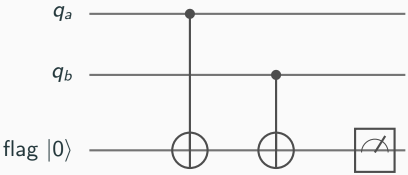
a flip at qubit 2, copied down its whole subtree

- A CNOT copies bit flips:
  - $\mathrm{CNOT}\,(X \otimes I) = (X \otimes X)\,\mathrm{CNOT}.$
- Damage depends on when the fault happens, not how big it was:

fault at	flipped w of $N = 15$	frequency $N - 2w$
depth 1	7	1
depth 2	3	9
a leaf	1	13

The frequency is a headcount: what the certificate reads.

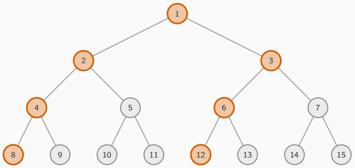
What a flag check measures, and what it does not



	$q_a q_b$	flag reads
branch $\|0 \cdots 0\rangle$	00	0
branch $\|1 \cdots 1\rangle$	11	0
after a flip on q_a	10	1
after a flip on *both*	11	0 (missed)

- Same answer on both branches: the flag learns nothing about *which*, so the superposition survives.
- Flag reads 1: odd bit-flip parity in its region; discard the shot.
- The missed double flip (two independent faults) moves the frequency to $N - 4$: the certificate catches what the flag cannot.

One check watches half the tree



one check on the pair $(8, 12)$: the paths back to their common ancestor

checks	qubits watched	coverage
$(8, 12)$	7 of 15	47%
$+(10, 14)$	11 of 15	73%

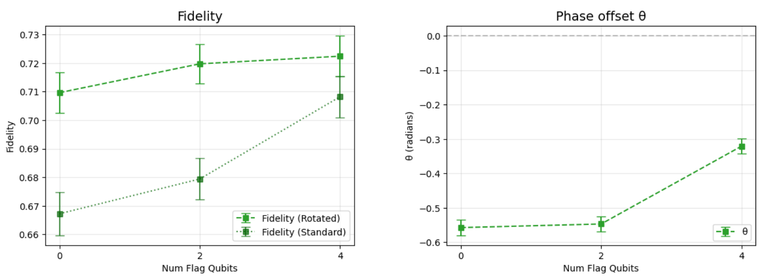
- The check compares two qubits, so a fault is visible only if it reaches exactly one of them: true on these paths, not at the split (both, cancels) or off them (neither). Coverage counts that path, a proxy rather than a guarantee.
- Each extra check adds less than the last (7, then 4): submodular, so greedy is within $1 - 1/e$ of optimal.
- Cost: acceptance, and extra gates can hurt fidelity. A hardware-dependent break-even.

Adapted from IBM's low-overhead error detection[2,3] to trapped-ion connectivity.

[2] Martiel, Simon, and Ali Javadi-Abhari. "Low-overhead error detection with spacetime codes." arXiv:2504.15725 (2025).

[3] Javadi-Abhari, Ali, et al. "Big cats: entanglement in 120 qubits and beyond." arXiv:2510.09520 (2025).

Quantinuum H2-1 hardware, 50 qubits



- 19 random angles instead of $2N = 100$: $5\times$ fewer circuits; 1000 shots per circuit (cost-limited), hence the wide error bars.
- $F_{\mathrm{rot}} = (P + C)/2$ scores against the GHZ state with the measured phase folded in; the standard fidelity pays $\cos\theta$. Rotated fidelity ≈ 0.72 with 4 flags; both improve with flags.
- Large phase offset at zero flags: consistent with accumulated coherent calibration error, amplified by $\theta = \sum_i \epsilon_i$ at $N = 50$.

Run it yourself: metriq-gym

```
$ pip install metriq-gym
$ mgym job dispatch ghz.example.json -p quantinuum -d H2-1E
Dispatching GHZ (N = 26, compressed sensing, M = 17 angles)...
job id 7f3a91c2 submitted
$ mgym job poll latest
population = 0.951   coherence = 0.906   n_rec = 26 ✓
fidelity (rotated) = 0.93
```

- One dispatch, one poll, on any supported backend (IBM, Quantinuum, IonQ, local simulators); the estimator and the $n_{\mathrm{rec}} = N$ size check come built in.
- DFE and full parity oscillation are included as alternative verification methods for comparison.

`github.com/unitaryfoundation/metriq-gym`

Thanks!

Vincent Russo

vprusso.github.io
vincentrusso1@gmail.com
github.com/vprusso

Paper: arXiv:2604.27824

interactive figures + blog post